Lie Algebras and their Representations

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Michaelmas 2023*

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Lecture 1

1 Introduction

A *Lie group* is fundamentally a group, which also a (smooth) manifold. For example, $\mathrm{GL}_n, \mathrm{SL}_n, \mathrm{SO}_n, \mathrm{Sp}_{2n}$.

> **Example 1.1**
> A prototypical example of a Lie group is the circle group S^1.

Let G be a Lie group. Then the *Lie algebra* of G is the tangent space at the identity e of G. That is,

$$\mathfrak{g} = \mathrm{T}_e G$$

$\mathfrak{g}$ is a vector space, with additional structure, which we will see later.

By taking a derivative, we turn the conjugation map

$$\begin{aligned} G &\to \mathrm{Aut}(G) \\ g &\mapsto g(\cdot)g^{-1} \end{aligned}$$

into a map

$$\mathrm{ad} : \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$$

called the *adjoint*. This gives a bilinear map

$$\begin{aligned} [\cdot,\cdot] : \mathfrak{g} \times \mathfrak{g} &\to \mathfrak{g} \\ [x, y] &= \mathrm{ad}(x)(y) \end{aligned}$$

> **Example 1.2**
> If $G = \mathrm{GL}_n(\mathbb{R})$, then we have that $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{R}) = \mathrm{Mat}_n(\mathbb{R})$, with
>
> $$[x, y] = xy - yx$$

What do Lie algebras tell us about the structure of the Lie group G?

- We will define the *root system* of $\mathfrak{g}$, and this then tells us about commutator relations in G (see Carter's book).

*Based on lectures by Stuart Martin. Last updated October 6, 2023.

- We will define the *Weyl group* of $\mathfrak{g}$. For example, the Weyl group of $\mathfrak{gl}_n(\mathbb{C})$ is isomorphic to S_n, there is an embedding $S_n \hookrightarrow \mathrm{GL}_n(\mathbb{C})$, vie permutation matrices. Let B denote the Borel subgroup of upper triangular matrices in $\mathrm{GL}_n(\mathbb{C})$, then there exists a *Bruhat decomposition*

$$G = \bigsqcup_{w \in S_n} BwB$$

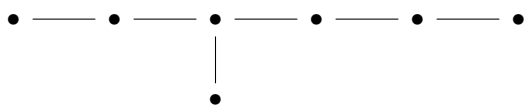
Lie algebras also give us information of the representation theory of G and $\mathfrak{g}$. For example, there exists a bijection

$$\{\text{finite dimensional } \mathbb{C}\text{-representations of } \mathrm{SL}_n(\mathbb{C})\} \leftrightarrow \{\text{finite dimensional } \mathbb{C}\text{-representations of } \mathfrak{sl}_n(\mathbb{C})\}$$

Moreover, we can describe the right hand side completely.

In addition, Lie algebras have applications in Algebraic Geometry, for example, we can use Lie algebras to build families of surfaces, of equivalenrly, algebraic curves (See book by Slodovy).

We will define the *Dynkin diagrams* of *semisimple Lie algebras*. For example,



is a Dynkin diagram of type E_7, and understanding the Dynkin diagram tells us about the singularities on the surfaces.

Moreover, Lie algebras also have applications in number theory, root systems/Weyl groups give the structure of groups over $\mathbb{Q}_p$, the p-adic integers (see paper on Moodle). Local Langlands correspondence predicts a relationship

$$\{\text{Galois theory of local fields}\} \leftrightarrow \{\text{complex Lie theory}\}$$

Finally, there any other applications, for example algebraic groups, quantum groups, theoretical physics, quantum mechanics.

2 Basic definitions and examples

Let k be a field. Most of the time, $k = \mathbb{C}$, but not always. We will sometimes point out how things can go wrong in characteristic p.

Definition 2.1 (Lie algebra)

A *Lie algebra* over k is a vector space $\mathfrak{g}$ over k, together with a bilinear pairing

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$$

satisfying

1. $[xx] = 0$ for all $x \in \mathfrak{g}$,
2. the Jacobi identity
$$[x[yz]] + [y[zx]] + [z[xy]] = 0$$

Notation 2.2. Note that when clear, we will write $[xy] := [x, y]$.

Remark 2.3. In particular, we have antisymmetry, i.e.

$$[xy] = -[yx]$$

Definition 2.4 ((Lie) subalgebra)

A k-vector subspace $\mathfrak{h}$ of $\mathfrak{g}$ is a *(Lie) subalgebra* if $\mathfrak{h}$ is closed under the Lie bracket of $\mathfrak{g}$. That is, for all $x, y \in \mathfrak{h}$, $[xy] \in \mathfrak{h}$.

Example 2.5

Let V be a finite dimensional k-vector space, then

1. Let $\mathfrak{gl}(V) = \mathrm{End}(V)$, with $[xy] = xy - yx$. If we choose a basis for V, then we can identify $\mathfrak{gl}(V)$ with $\mathrm{Mat}_n(k)$. In this case, we will write $\mathfrak{gl}_n$ or $\mathfrak{gl}_n(k)$.
2. Let
$$\mathfrak{sl}(V) = \{x \in \mathfrak{gl}(V) \mid \mathrm{tr}(x) = 0\}$$
This defines a subalgebra of $\mathfrak{gl}(V)$, called the *special linear Lie algebra*, with $\dim(\mathfrak{sl}(V)) = \dim(V)^2 - 1$. The standard basis of $\mathfrak{sl}(V)$ is given by
$$E_{i,j} \text{ for } i \neq j$$
$$E_{i,i} - E_{i+1,i+1}$$
We will often write $\mathfrak{sl}_n$ for this Lie algebra.

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